ACM 104 HW #3

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1. Inner Products vs Norms

a) We can use the polarization identity to recover the inner product from the norm.

Proof:
$$\|u+v\|^2 = \langle u+v, u+v\rangle = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$$
$$\|u-v\|^2 = \langle u-v, u-v\rangle = \|u\|^2 - 2\langle u, v\rangle + \|v\|^2$$

Subtracting and simplifying:
$$\frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2\right) = \langle u, v\rangle$$

$$\boxed{\text{Yes, it's possible.}}$$

b) Suppose $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are two inner products that have the same norm i.e. $\|\cdot\|_1 = \|\cdot\|_2$

Then for all $v \in V$:
$$\|v\|_1^2 = \langle v, v\rangle_1 \text{ and } \|v\|_2^2 = \langle v, v\rangle_2$$

By polarization identity:
$$\langle u, v\rangle_1 = \frac{1}{4}\left(\|u+v\|_1^2 - \|u-v\|_1^2\right)$$
$$\langle u, v\rangle_2 = \frac{1}{4}\left(\|u+v\|_1^2 - \|u-v\|_1^2\right)$$

Since $\|\cdot\|_1 = \|\cdot\|_2$ by assumption:
$$\langle u, v\rangle_1 = \langle u, v\rangle_2 \quad \forall\, u, v \in V$$

Therefore $\langle \cdot, \cdot \rangle_1 = \langle \cdot, \cdot \rangle_2$, contradicting the assumption of distinctiveness.

$$\boxed{\text{No}}$$

2. Inner Products on Function Spaces

a) $\boxed{\langle f, g\rangle_2 \text{ is the inner product}}$

$\langle f, g\rangle_1$ fails positive definiteness:

If we take some function $f(x) = c$, $c \neq 0$:

$$\langle f, f\rangle_1 = \int_0^1 (f'(x))^2 dx = \int_0^1 0\, dx = 0$$

Since $\langle f, f\rangle_1 = 0$, but $f \neq 0$, this violates positive definiteness.

b) $\langle f, g\rangle = \int_0^1 (f(x)g(x) + f'(x)g'(x))dx$

$$\|f\| = \sqrt{\int_0^1 (f(x)^2 + (f'(x))^2)\, dx}$$

Cauchy-Schwarty:

$$\left|\int_0^1 (f(x)g(x) + f'(x)g'(x))dx\right| \leq \sqrt{\int_0^1 (f(x)^2 + (f'(x))^2 dx} \cdot \sqrt{\int_0^1 (g(x)^2 + (g'(x))^2) dx}$$

Triangle inequality:

$$\sqrt{\int_0^1 ((f+g)^2 + (f'+g')^2) dx} \leq \sqrt{\int_0^1 (f^2 + (f')^2) dx} \cdot \sqrt{\int_0^1 (g^2 + (g')^2) dx}$$

c) $f(x) = 1$, $f'(x) = 0$; $g(x) = e^x$, $g'(x) = e^x$

$$\langle f, g\rangle = \int_0^1 (1 \cdot e^x + 0 \cdot e^x) dx = e - 1$$

$$\|f\| = \int_0^1 (1 + 0)\, dx = 1$$

$$\|g\| = \int_0^1 (e^{2x} + e^{2x}) dx = e^2 - 1$$

$$\theta = \cos^{-1}\left(\frac{\langle f, g\rangle}{\|f\| \cdot \|g\|}\right) = \cos^{-1}\left(\frac{e-1}{\sqrt{e^2-1}}\right) = \boxed{47.2^\circ \text{ or } 0.823 \text{ rad}}$$

d) We can set up Cauchy-Schwartz and use this expression:

$$\left|\int_a^b f(x)\cdot 1\,dx\right| \leq \sqrt{\int_0^b f(x)^2\,dx} \cdot \sqrt{\underbrace{\int_a^b 1^2\,dx}_{b-a}}$$

$$\left(\int_a^b f(x)\,dx\right)^2 \leq (b-a)\int_a^b f(x)^2\,dx$$

$$\boxed{\left(\frac{1}{b-a}\int_a^b f(x)\,dx\right)^2 \leq \frac{1}{b-a}\int_a^b f(x)^2\,dx}$$

3. The K-Means Algorithm For Clustering

See page 6 of this document.

4. Gram Matrices $\qquad \langle f,g\rangle = \int_0^1 f(x)g(x)\,dx$

a)

$$G = \begin{bmatrix} \langle 1,1\rangle & \langle 1,e^x\rangle & \langle 1,e^{2x}\rangle \\ \langle e^x,1\rangle & \langle e^x,e^x\rangle & \langle e^x,e^{2x}\rangle \\ \langle e^{2x},1\rangle & \langle e^{2x},e^x\rangle & \langle e^{2x},e^{2x}\rangle \end{bmatrix} = \begin{bmatrix} 1 & e-1 & e^2-1 \\ e-1 & \frac{e^2-1}{2} & \frac{e^3-1}{3} \\ \frac{e^2-1}{2} & \frac{e^3-1}{3} & \frac{e^4-1}{4} \end{bmatrix}$$

b) $\boxed{\text{Yes, positive definite}}$ For $c = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$,

$$f(x) = c_1 \cdot 1 + c_2 e^x + c_3 e^{2x}$$

Then:

$$c^T G c = \langle f, f\rangle = \int_0^1 f(x)^2\,dx$$

Since $1, e^x, e^{2x}$ are lin. indep., $f(x) \not\equiv 0$ whenever $c \neq 0$

For non zero continuous function f, $f(x)^2 > 0$, meaning:

$$\int_0^1 f(x)^2\,dx > 0$$

Therefore, $c^T G c > 0$ for all $c \neq 0$, so G positive definite.

c) $\langle f,g\rangle_2 = \int_0^1 (fg + f'g')\,dx$; $(e^x)' = e^x$, $(e^{2x})' = 2e^{2x}$

$\langle 1,1\rangle_2 = 1$, $\langle 1,e^x\rangle_2 = \int_0^1 e^x\,dx = e-1$, $\langle 1,e^{2x}\rangle_2 = \int_0^1 e^{2x}\,dx = \frac{e^2-1}{2}$

$\langle e^x,e^x\rangle_2 = \int_0^1 (e^{2x}+e^{2x})\,dx = e^2-1$

$\langle e^x,e^{2x}\rangle_2 = e^3-1$; $\langle e^{2x},e^{2x}\rangle_2 = \frac{5}{4}(e^4-1)$

$$G_2 = \begin{bmatrix} 1 & e-1 & \frac{e^2-1}{2} \\ e-1 & e^2-1 & e^3-1 \\ \frac{e^2-1}{2} & e^3-1 & \frac{5}{4}(e^4-1) \end{bmatrix}$$

G_2 is positive definite because set is lin. indep. and because $\langle\cdot,\cdot\rangle_2$ is an inner product.

d) $\boxed{\text{No}}$

The positive definity of a Gram matrix depends only on the linear independence of the vectors and not on the inner product.

For any inner product and any set of vectors $v_1, \ldots, v_n$:
The Gram matrix is always positive semi-definite and is positive definite iff the vectors are lin. indep.

Since linear independence is a natural product of vectors:
If G_1 is positive definite $\longleftrightarrow$ vectors are lin. indep $\longleftrightarrow$ G_2 is positive definite.

ACM/IDS 104 - Problem Set 3 - MATLAB Problems

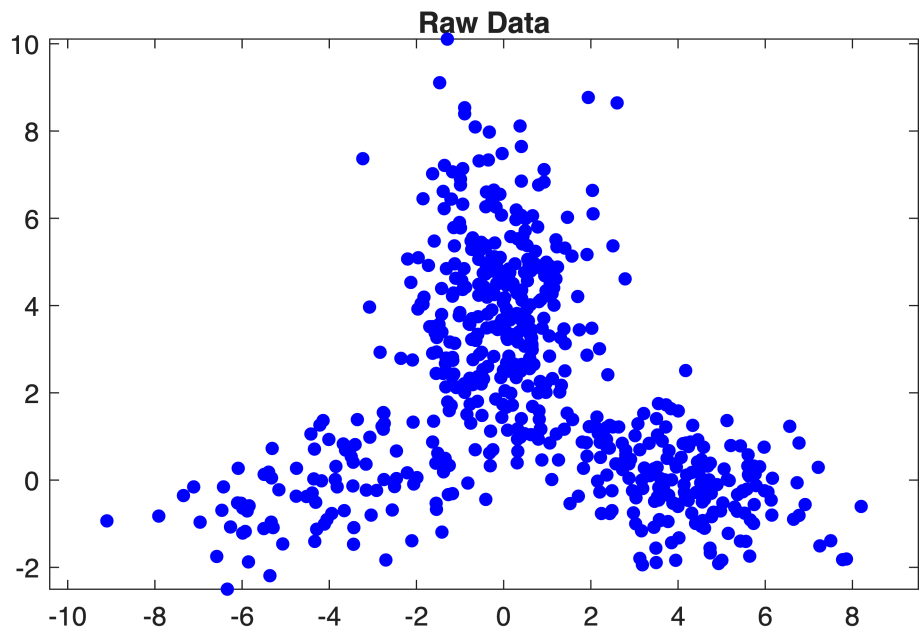
Before writing your MATLAB code, it is always good practice to get rid of any leftover variables and figures from previous scripts.

```
clc; clear; close all;
```

Problem 3 (10 points) The K-Means Algorithm for Clustering

In this problem, we will be implementing the K-Means Algorithm for clustering vectors in $\mathbb{R}^2$. To start, let us load and visualize our dataset.

```
load clustering_data.mat
figure;
plot(x(:,1),x(:,2),'.b','MarkerSize',15);
title("Raw Data")
axis equal;
```




```
rng(2016); % for reproducibility
```

Part (a) The Algorithm

In this part, we will implement the algorithm for $K = 3$, which seems to be a reasonable number of clusters for this dataset. Show the clusters after the 1st, 5th, 10th and last iteration. This will illustrate the cluster evolution as the algorithm runs. `subplot()` is very useful here.

To further aid our understanding of the algorithm, we will be keeping track of a few things:

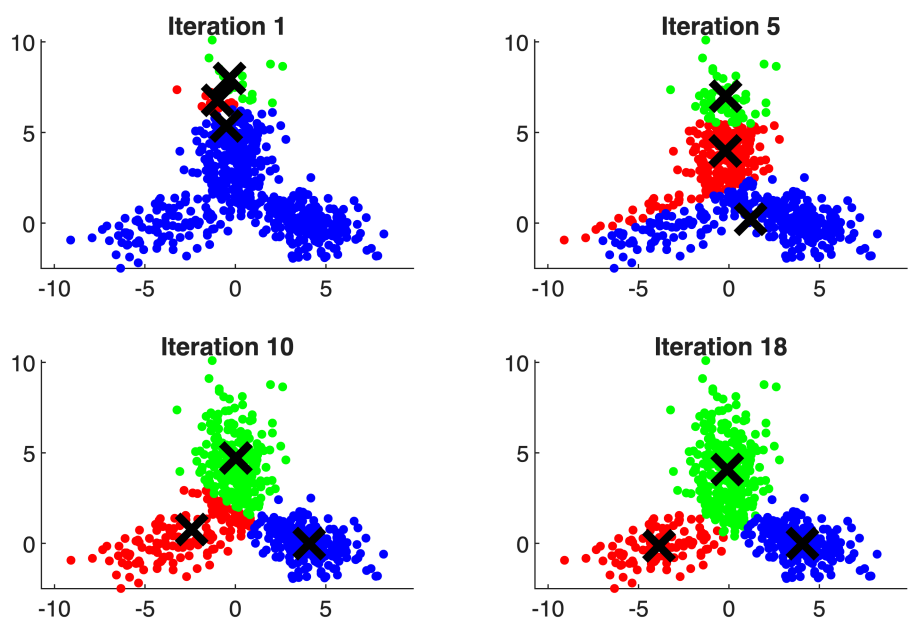
1. Keep track of the evolution of the representatives for each cluster. You can do so by updating the variable `z` after each iteration.
2. Keep track of the objective function value at each iteration.

Once you are done implementing the algorithm, feel free to change the value of `K` and see how the clusterings are altered.

```
%{
Some useful setup done for you
%}
K = 3; % number of clusters
n = length(x); % number of vectors
c = zeros(n, 1); % clustering : c(i) is the cluster number for vector
x(i, :)
iter = 0; % iteration number
z = datasample(x, K); % initial representatives

z_history = zeros(K, 2, 100);
z_history(:, :, 1) = z;
obj_func = zeros(100, 1);

%{
Step 1: Partitioning into clusters given our initial representatives
%}
for i = 1:n
    distances = zeros(K, 1);
    for k = 1:K
        distances(k) = norm(x(i,:) - z(k,:));
    end
    [~, c(i)] = min(distances);
end

%{
Increment the iteration number
Update the objective function
%}
iter = iter + 1;
obj_func(iter) = 0;
for i = 1:n
    obj_func(iter) = obj_func(iter) + norm(x(i,:) - z(c(i),:))^2;
end

c_at_iter = zeros(n, 4); % store c at 1, 5, 10, and final one
z_at_iter = zeros(K, 2, 4);
c_at_iter(:, 1) = c;
z_at_iter(:, :, 1) = z;

%{
```

```
Iterate until convergence:
%}
converged = false;
while ~converged
    % compute mean of each cluster
    z_old = z;
    for k = 1:K
        cluster_points = x(c == k, :);
        if ~isempty(cluster_points)
            z(k, :) = mean(cluster_points, 1);
        end
    end

    % resssign to nearest representative
    for i = 1:n
        distances = zeros(K, 1);
        for k = 1:K
            distances(k) = norm(x(i,:) - z(k,:));
        end
        [~, c(i)] = min(distances);
    end
    iter = iter + 1;
    z_history(:, :, iter) = z;
    obj_func(iter) = 0;
    for i = 1:n
        obj_func(iter) = obj_func(iter) + norm(x(i,:) - z(c(i),:))^2;
    end

    % store at specific iterations
    if iter == 5
        c_at_iter(:, 2) = c;
        z_at_iter(:, :, 2) = z;
    elseif iter == 10
        c_at_iter(:, 3) = c;
        z_at_iter(:, :, 3) = z;
    end

    % check convergence
    if norm(z - z_old) < 1e-6 || iter > 100
        converged = true;
        c_at_iter(:, 4) = c;
        z_at_iter(:, :, 4) = z;
    end
end

z_history = z_history(:, :, 1:iter);
obj_func = obj_func(1:iter);

%{
Plotting
```

```
%}
figure;
colors = ['r', 'g', 'b'];
iterations_to_show = [1, 5, 10, iter];

for idx = 1:4
    subplot(2, 2, idx);
    hold on;
    for k = 1:K
        cluster_idx = (c_at_iter(:, idx) == k);
        plot(x(cluster_idx, 1), x(cluster_idx, 2), '.', 'Color', colors(k),
'MarkerSize', 10);
    end
    plot(z_at_iter(:, 1, idx), z_at_iter(:, 2, idx), 'kx', 'MarkerSize',
15, 'LineWidth', 3);

    title(sprintf('Iteration %d', iterations_to_show(idx)));
    axis equal;
    hold off;
end
```

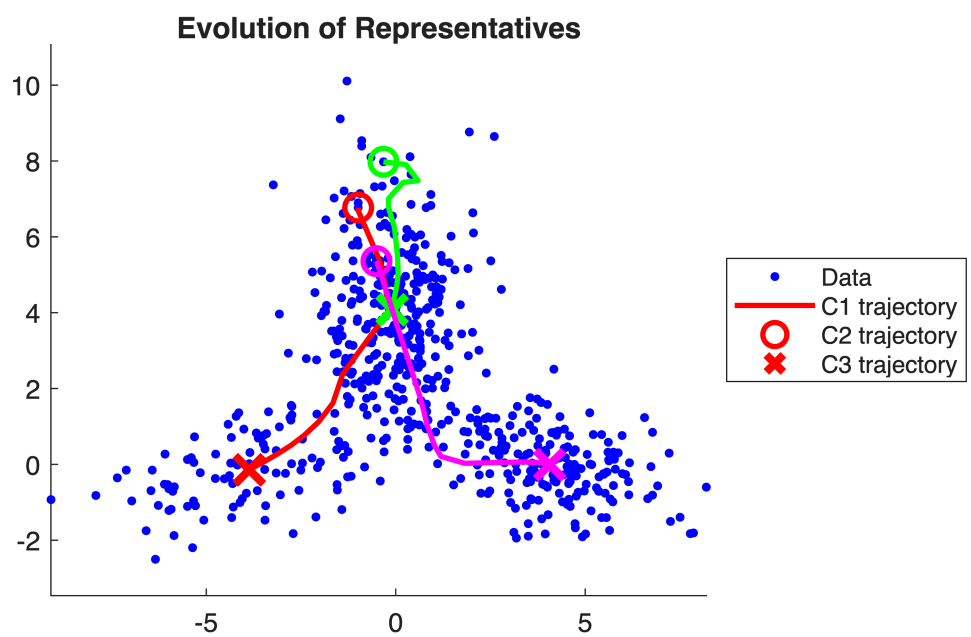


Part (b) Evolution of Representatives

Now, let us see the dynamics of the representatives. Plot the original (raw) data and the trajectories of the cluster representatives in one figure. `squeeze()` might be useful here.

```
figure;
hold on;

plot(x(:,1), x(:,2), '.b', 'MarkerSize', 10);
colors = ['r', 'g', 'm'];
```

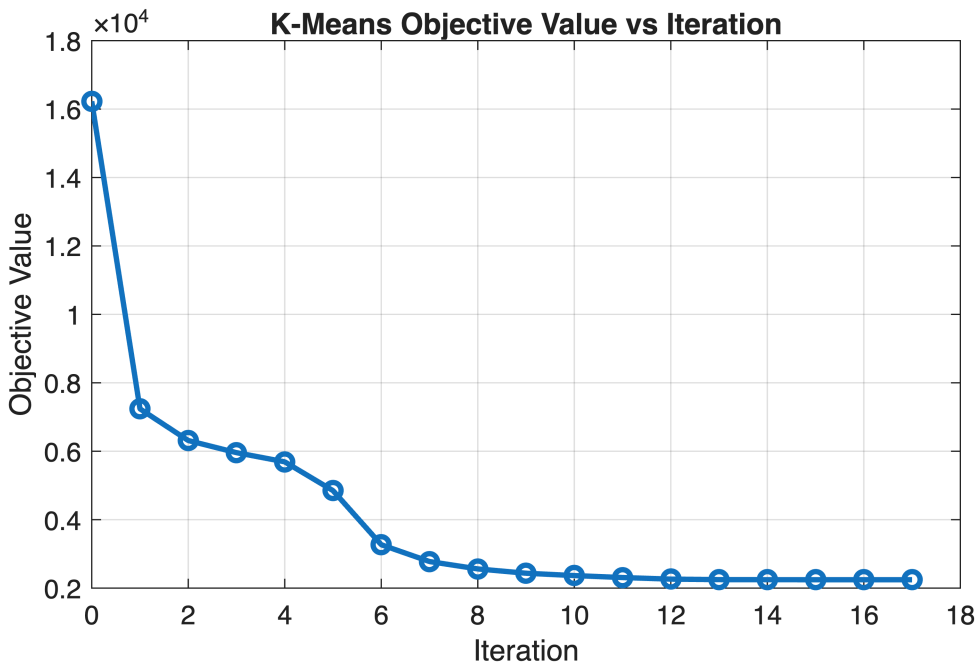
```
for k = 1:K
    trajectory = squeeze(z_history(k, :, :));
    plot(trajectory(1, :),trajectory(2, :), '-', 'Color', colors(k),
'LineWidth', 2);
    plot(trajectory(1, 1),trajectory(2, 1), 'o', 'Color', colors(k),
'MarkerSize', 10, 'LineWidth', 2);
    plot(trajectory(1, end),trajectory(2, end), 'x', 'Color', colors(k),
'MarkerSize', 15, 'LineWidth', 3);
end

title('Evolution of Representatives');
legend('Data', 'C1 trajectory', 'C2 trajectory', 'C3 trajectory',
'Location', 'eastoutside'); % key was overlapping so I moved it to side
axis equal;
hold off;
```



Part (c) Objective Function

Plot the values of the objective function against the iteration number. Does the trend match what we expect? Report the minimum value of the objective function using `disp()`.

```
figure;
plot(0:iter-1, obj_func(1:iter), '-o', 'LineWidth', 2);
xlabel('Iteration');
ylabel('Objective Value');
title('K-Means Objective Value vs Iteration');
grid on;
```




```
disp('Minimum value of objective function:');
```

```
Minimum value of objective function:
```

```
disp(min(obj_func(1:iter)));
```

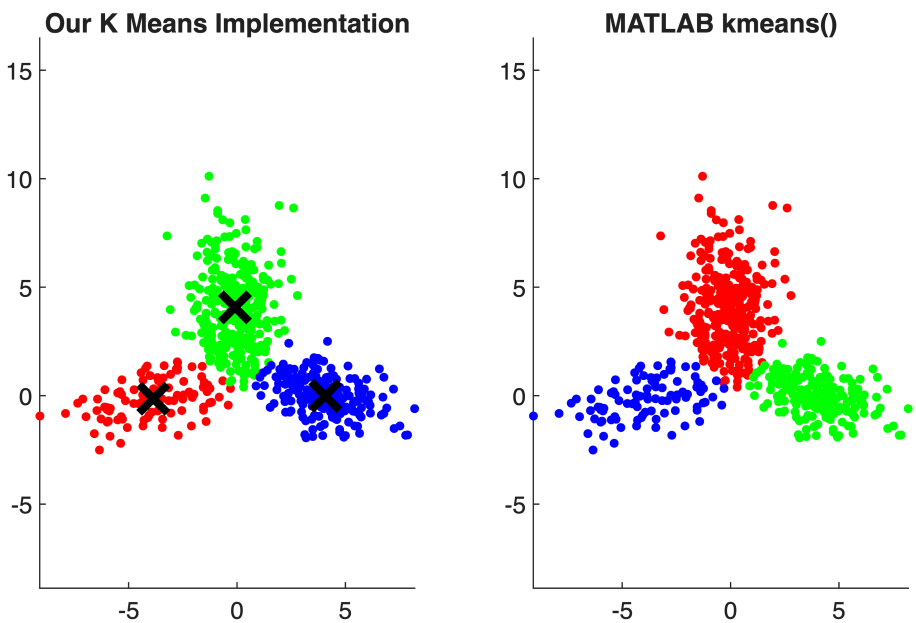
```
2.2452e+03
```

Part (d) Comparing with `kmeans()`

The K-means algorithm is a very popular clustering algorithm and it is often used in various applications. As such, it is implemented in most numerical software packages. In MATLAB, it exists as the built-in function `kmeans()`. In the simplest form, `idx = kmeans(X, k)` performs K-means clustering to partition the rows in the data matrix `X` into `k` clusters, and returns a vector `idx` containing cluster indices for each row. To compare your implementation with the built-in one, plot the clustering your obtained in (a) and the clustering obtained by using `kmeans()`.

```
rng(2016); % for reproducibility
idx_builtin = kmeans(x, K);

figure;

% Our Implementation
subplot(1, 2, 1);
hold on;
colors = ['r','g','b'];
for k = 1:K
    cluster_idx = (c == k);
    plot(x(cluster_idx, 1), x(cluster_idx, 2), '.', 'Color', colors(k),
'MarkerSize', 10);
```

```
end
plot(z(:, 1), z(:, 2), 'kx', 'MarkerSize', 15, 'LineWidth', 3);
title('Our K Means Implementation');
axis equal;
hold off;

% built-in kmeans
subplot(1, 2, 2);
hold on;
for k = 1:K
    cluster_idx = (idx_builtin == k);
    plot(x(cluster_idx, 1), x(cluster_idx, 2), '.', 'Color', colors(k),
'MarkerSize', 10);
end
title('MATLAB kmeans()');
axis equal;
hold off;
```



Problem 5 (10 points) Gram Matrices for Text Classification

Gram matrices are used in many different applications. Here is a simple example of application to classification of text documents. Suppose we have a collection of n text documents and we want to measure similarity between them. First, we predefine a list of words (the "dictionary") $w_1, w_2, w_3, \cdots, w_m$ appearing in the documents. Then, for each text i we construct a vector $x_i \in \mathbb{R}^m$ whose j^{th} component is the number of times the word w_j appears in the text. This way, each text is represented as a vector in $\mathbb{R}^m$. This representation is often referred to as the "bag of words" representation.

Let $\langle \cdot, \cdot \rangle$ be the usual dot product in $\mathbb{R}^m$ and $\| \cdot \|$ be the Euclidean norm. Let us normalize all vectors $z_i = \frac{x_i}{\|x_i\|}$ and define:

$$Z = [z_1 \cdots z_n] \in \mathbb{M}_{m \times n}$$

Let G be the Gram matrix associated with $z_1 \cdots z_n$:

$$G = Z^T Z \in \mathbb{M}_{n \times n}$$

Since all z_i are unit vectors:

$$G_{ij} = z_i^T z_j = \|z_i\| \|z_j\| \cos \theta_{ij} = \cos \theta_{ij}$$

where $\theta_{ij}$is the angle between z_i and z_j, which of course is the same as the angle between x_i and x_j. This is widely use as a *measure of similarity* between documents. Hence, the more similar two texts i and j are, the closer G_{ij} is to 1.

Now, let us apply this method to real texts. The task is to compare the text of the US Constitution (found here) with the Wikipedia pages of Bernie Sanders, Hillary Clinton, Donald Trump, Ted Cruz, and John Kasich, and to find the page which is "closest" to the Constitution.

Part (a) Setup

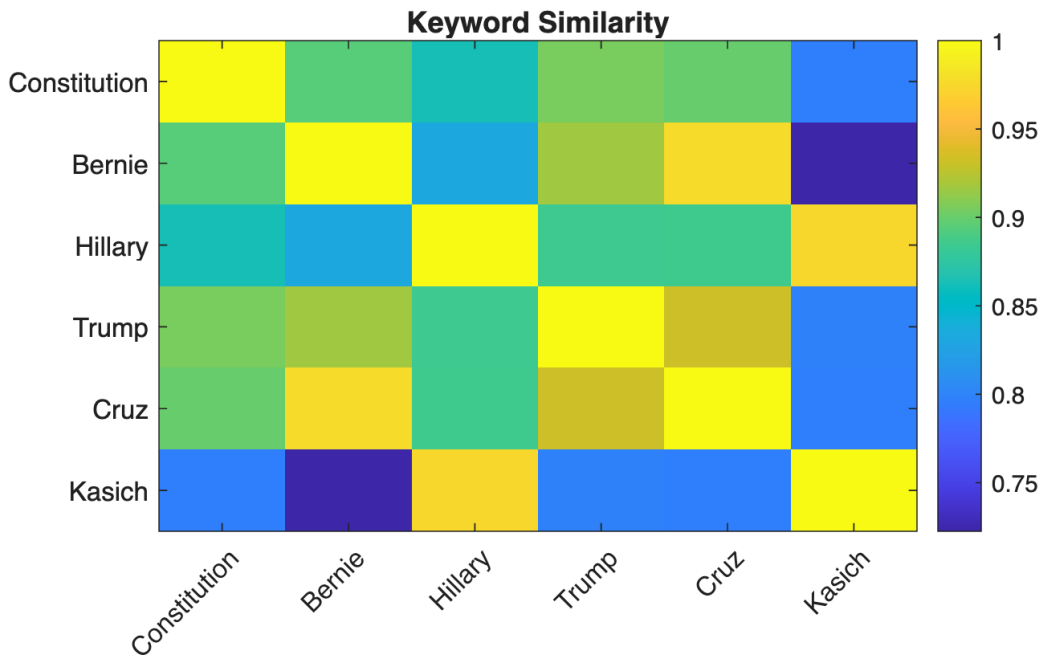
To start, we will define a list of $m = 10$ words. To find the most "relevant words", copy the text of the Constitution and paste it into a word-cloud generator (such as this one). High frequency words appear large and we can form the dictionary from these words. Note that common sense is needed: words like "amendment" and "section" would appear large, but it makes no sense to include them into the dictionary. Also, it makes sense to include "United States" as a single "word", not two separate words, etc. But you are free to chose your own dictionary. In the code box below, please fill out the dictionary with your chosen words.

```
%{
DICTIONARY:
1) States
2) President
3) United
4) State
5) any
6) Congress
7) Section
8) all
9) Law
10) Office
%}
```

Part (b) Fun

With your specified dictionary, let us see which Wikipedia page is closest. First, construct $n = 6$ vectors (as specified above), one for each Wikipedia page and one for the Constitution. No need to get fancy here; just go to the related page and search for the number of occureneces of the words in your dictionary. Put all your results in a column vector for each candidate:

```
%{
Report your results here
%}
% I used a python BeautifulSoup script to fetch these given the urls
% see bottom of document
constitution = [128; 120; 85; 80; 78; 64; 55; 42; 39; 37];
bernie = [102; 53; 101; 34; 15; 47; 1; 33; 23; 29];
hillary = [78; 75; 76; 126; 13; 19; 1; 19; 35; 27];
trump = [89; 109; 92; 53; 16; 24; 1; 29; 50; 67];
cruz = [92; 50; 81; 44; 10; 35; 0; 21; 46; 19];
kasich = [38; 40; 28; 89; 5; 15; 0; 18; 25; 15];

%{
Normalize your vectors
%}
X = [constitution, bernie, hillary, trump, cruz, kasich];
Z = zeros(size(X));
for i = 1:size(X, 2)
    Z(:, i) = X(:, i) / norm(X(:, i));
end

%{
Construct the Gram matrix G as discussed
%}
G = Z' * Z;

%{
You can visualize G by uncommenting the following lines of code
%}
figure;
imagesc(G);
colorbar;
title('Keyword Similarity');
labels = {'Constitution', 'Bernie', 'Hillary', 'Trump', 'Cruz', 'Kasich'};
set(gca, 'XTick', 1:6, 'XTickLabel', labels, 'YTick', 1:6, 'YTickLabel',
labels);
xtickangle(45);
```




```
%{
Find the page closest to the Constitution text and report your
answer using disp()
%}
similarities = G(1, 2:end);
[max_similarity, max_idx] = max(similarities);
pages = {'Bernie', 'Hillary', 'Trump', 'Cruz', 'Kasich'};

disp('Similarity scores with the Constitution:');
```

```
Similarity scores with the Constitution:
```

```
for i = 1:length(pages)
    disp([pages{i}, ': ', num2str(similarities(i))]);
end
```

```
Bernie: 0.8944
Hillary: 0.86287
Trump: 0.905
Cruz: 0.90008
Kasich: 0.79609
```

```
disp(' ');
```

```
disp(['The Wikipedia page closest to the Constitution is: ',
pages{max_idx}]);
```

```
The Wikipedia page closest to the Constitution is: Trump
```

```
disp(['Similarity score: ', num2str(max_similarity)]);
```

```
Similarity score: 0.905
```

```
%{
# Web Parser Keyword Script
import requests
from bs4 import BeautifulSoup
from collections import Counter
import re

target_words = ["States", "President", "United", "State", "any",
"Congress", "Section", "all", "Law", "Office"]

def count_words(url):
    # server blocks automated scripts so we use spoofing
    headers = {
        "User-Agent": (
            "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) "
            "AppleWebKit/537.36 (KHTML, like Gecko) "
            "Chrome/118.0.0.0 Safari/537.36"
        )
    }
    try:
        response = requests.get(url, headers=headers)
        response.raise_for_status()
        soup = BeautifulSoup(response.text, 'html.parser')
        text = soup.get_text(separator=' ')
        words = re.findall(r'\b\w+\b', text.lower())
        counts = Counter(words)
        return [counts[word.lower()] for word in target_words]
    except requests.RequestException as e:
        print(f"Error fetching {url}: {e}")
        return [0] * len(target_words)

if __name__ == "__main__":
    urls = []
    print("Gimme urls:")
    for i in range(6):
        urls.append(input(f"URL {i+1}: ").strip())

    print("\nResults:\n")
    labels = ['a', 'b', 'c', 'd', 'e', 'f']
    for label, url in zip(labels, urls):
        counts = count_words(url)
        formatted = "; ".join(map(str, counts))
        print(f"{label} = [{formatted}];")
%}
```

Submit the obtained result to the corresponding piazza poll. Remember: this is not about your political preferences, it is about your data analysis :)